

# Derrick Chen

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## EDUCATION

### Georgia Institute of Technology

Expected December 2027

*Bachelor of Science in Computer Science*

- Relevant Coursework: Data Structures and Algorithms, Artificial Intelligence, Object Oriented Programming, Computer Architecture, Machine Learning, Operating Systems, Database Design

## TECHNICAL SKILLS

**Languages:** Java, Python, JavaScript, TypeScript, C, C++

**Frameworks:** React, Node.js, Next.js, Flask, Django, Spring Boot, Express

**Developer Tools:** AWS, Git/GitHub, Vercel, Jira, Playwright, Azure, PyTest, Docker, Kafka

**Databases:** PostgreSQL, MongoDB, MySQL

**AI Development:** Model Context Protocol (MCP), AI Agents, Claude Code, Cursor, Prompt Engineering

## EXPERIENCE

### Software Engineer Co-Op

Fall 2026

*IBM*

San Jose, CA

- Incoming for Fall 2026 cohort (AI compliance at scale)

### Software Engineer Intern

June 2026 - Aug 2026

*Walmart*

Bentonville, AR

- Designed an event-driven stocking confirmation flow processing **3,000+ requests/minute**, replacing inventory guesswork with real-time stocking updates saving a store hundreds of thousands annually
- Implemented the backend with Java, Spring Boot, and Apache Kafka, implementing event mapping, validation, and publishing logic with **90%+ code coverage**
- Built a Grafana and Prometheus/Apache JMeter dashboard tracking **p95/p99 latency**, HTTP throughput, Kafka publish rates, and CPU/heap utilization to monitor production metrics

### Software Engineer Intern

June 2025 - Aug 2025

*BigID*

Atlanta, GA

- Developed integration for **BigID customers** to stream DSPM cases into **AWS Security Lake** using **IAM** and **S3**, allowing customers to centralize DSPM data within their AWS environments
- Built the backend in **Node.js**, using ETL pipelines instead of RESTful APIs to handle terabytes of security data, cutting ingestion time by over **100x** (hours to minutes)
- Wrote **Jest unit tests** achieving **80%+ code coverage** and built end-to-end integration tests for DSPM - AWS workflows; used **Jenkins** to manage test environments and automate cloud deployments

### Undergraduate Researcher

Aug 2024 - Dec 2024

*GROWER Research Lab*

Atlanta, GA

- Wrote **Python** scripts to web scrape data on power outages from 13+ sources, expanding my team's dataset coverage by **15%** of U.S. states
- Cleaned and analyzed datasets with **1M+ records**, enabling data analysis in emergency response, public investments, and public policy
- Collaborated on a subteam of 6 using Agile methodologies such as daily standups, sprint planning, and pair programming, while partnering with Oak Ridge National Laboratory

## PROJECTS

### Referee Ruling Engine | Python, FastAPI, Next.js, TypeScript, PostgreSQL, pgvector

[ask-the-ref.com](https://ask-the-ref.com)

- Built an NBA rules verification engine averaging **1.6K users/visits monthly** that translates **natural language** basketball scenarios into a legal or illegal verdict
- Implemented a hybrid **RAG pipeline** combining **vector similarity search** and **full text search** to retrieve relevant sections from semantically chunked rules
- Developed a data ingestion pipeline to scrape, clean, and **hierarchically chunk** NBA rules by section, allowing retrieval via RAG
- Deployed a cartoon themed Next.js and TypeScript frontend, backed by a Python and FastAPI backend using a vector embedding model for semantic rule retrieval

## EXTRA

**Awards:** Coca-Cola First Generation Scholar, Zell Miller Scholarship Recipient

**Extracurriculars:** WebDev @ GT, Intramural Sports, Blockchain @ GT, Greek Life (Director of Career Development)